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Experiment: 1

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COURSE NAME: CSA4701 Deep Learning

1. Vector and Matrix Operations for Deep Learning

Scenario

A neural network receives input features representing a student's study hours, attendance, and assignment scores. Create vectors and matrices to represent the data and perform matrix multiplication to calculate weighted outputs

Algorithm

1. Start the program.
2. Import the NumPy and Matplotlib libraries.
3. Create a student input matrix containing study hours, attendance percentage, and assignment scores.
4. Normalize the study hours into a percentage scale.
5. Create a weight vector with weights of 0.30 for study hours, 0.30 for attendance, and 0.40 for assignment scores.
6. Perform matrix multiplication between the input matrix and the weight vector to calculate the weighted scores.
7. Display the input matrix, weight vector, and calculated weighted scores.
8. Create a summary table containing student details and their weighted scores.
9. Plot a bar chart to visualize the weighted scores of the students.
10. Display the plot with appropriate labels, title, and score values.

    Stop the program.

Source code

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt
```

```
# 1. Student input data
# Columns: Study Hours, Attendance (%), Assignment Score (%)
students = ["Student A", "Student B", "Student C"]

data = np.array([
   [8, 90, 85],
   [6, 75, 70],
   [10, 95, 95]
])

# 2. Create the input matrix
# Convert study hours out of 10 into a percentage
X = data.astype(float)
X[:, 0] = (X[:, 0] / 10) * 100

print("Input Matrix (Study Hours converted to %):")
print(X)

# 3. Create the weight vector
# Weights: Study Hours = 30%, Attendance = 30%,
# Assignment Score = 40%
weights = np.array([0.30, 0.30, 0.40])

print("\nWeight Vector:")
print(weights)

# 4. Matrix multiplication to calculate weighted scores
scores = X @ weights
```

```
print("\nWeighted Scores:")
print(scores)

# 5. Create a summary table
summary = pd.DataFrame({
    "Student": students,
    "Study Hours": data[:, 0],
    "Attendance (%)": data[:, 1],
    "Assignment Score (%)": data[:, 2],
    "Weighted Score (/100)": np.round(scores, 2)
})

print("\nSUMMARY TABLE")
print(summary.to_string(index=False))

# 6. Plot the weighted scores
plt.figure(figsize=(8, 5))
bars = plt.bar(students, scores)

plt.title("Weighted Student Scores Using Matrix Multiplication")
plt.xlabel("Students")
plt.ylabel("Weighted Score (out of 100)")
plt.ylim(0, 100)
plt.grid(axis="y", linestyle="--", alpha=0.4)

# Display score values above the bars
for bar, score in zip(bars, scores):
    plt.text(
        bar.get_x() + bar.get_width() / 2,
```

```
        bar.get_height() + 1,
        f"{score:.2f}",
        ha="center"
    )

plt.tight_layout()
plt.show()
```

Output

```
Input Matrix (Study Hours converted to %):
[[ 80.  90.  85.]
 [ 60.  75.  70.]
 [100.  95.  95.]]

Weight Vector:
[0.3 0.3 0.4]

Weighted Scores:
[85.  68.5 96.5]

SUMMARY TABLE
  Student  Study Hours  Attendance (%)  Assignment Score (%)  Weighted Score (/100)
Student A            8              90                    85                   85.0
Student B            6              75                    70                   68.5
Student C           10              95                    95                   96.5
```

